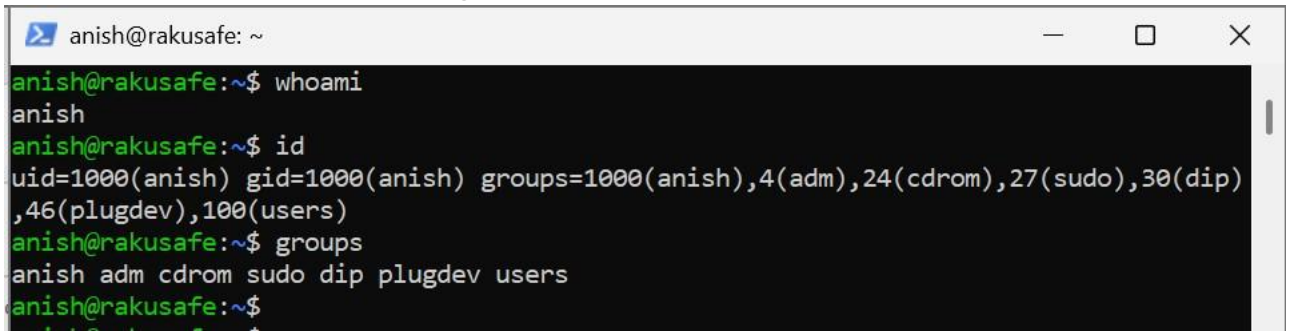
# Linux Fundamentals Day 4 — Interview-Oriented Machine Test

**All content preserved. Practical + Interview Questions.**

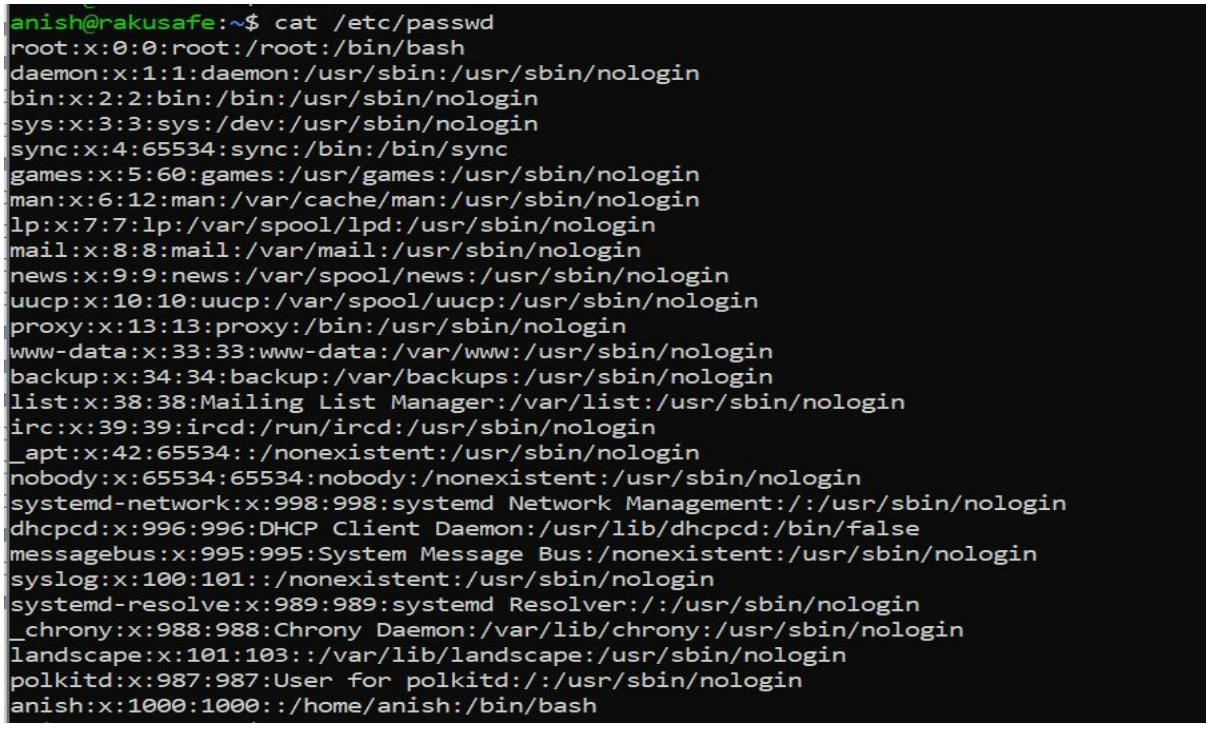
**Date: 01/10/2026**
**Submitted by: ANISHKRISHNAN**
**GitHub Repository**
**GitHub URL:**

### Task 1 — User Information Investigation




1. What is your current username? anish
2. Difference between UID and GID?
uid=1000(anish) → identifies the user
gid=1000(anish) → identifies the user's primary group
3. List all groups your user belongs to.
anish adm cdrom sudo dip plugdev users
4. Which group allows administrative access? Sudo

### Task 2 — Linux User Database



1.Which file stores Linux user information?
/etc/passwd
2. Explain ubuntu:x:1000:1000:Ubuntu User:/home/ubuntu:/bin/bash
ubuntu:x:1000:1000:Ubuntu User:/home/ubuntu:/bin/bash represents a Linux user account. It contains the username, password placeholder, UID, GID, user description, home directory, and login shell."
3. What is the purpose of the login shell field?
The login shell specifies the program that runs when a user logs in to Linux.
Example:/bin/bash

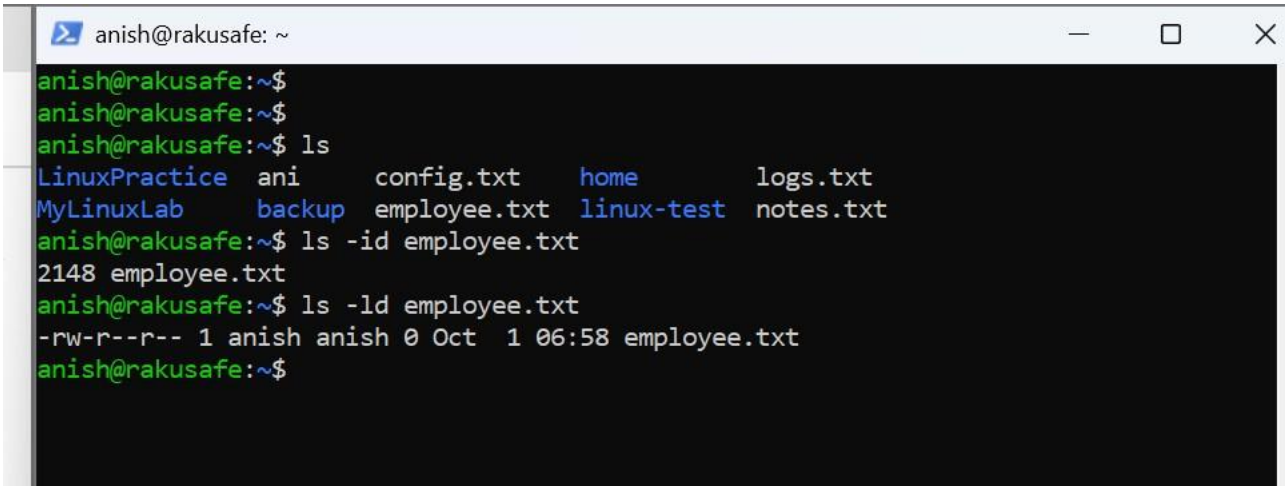
### Task 3 — Linux Group Database

```
anish@rakusafe:~$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,anish
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:
man:x:12:
proxy:x:13:
kmem:x:15:
dialout:x:20:
fax:x:21:
voice:x:22:
cdrom:x:24:anish
floppy:x:25:
tape:x:26:
sudo:x:27:anish
audio:x:29:
dip:x:30:anish
www-data:x:33:
backup:x:34:
operator:x:37:
list:x:38:
irc:x:39:
src:x:40:
shadow:x:42:
utmp:x:43:
video:x:44:
sasl:x:45:
plugdev:x:46:anish
staff:x:50:
```

1. Which file stores Linux group information? /etc/group
2. What is the purpose of a group in Linux?
The /etc/group file stores information about Linux groups and their members

3. Explain docker:x:999:ubuntu
“docker:x:999:ubuntu means the group name is docker, its GID is 999, and the user ubuntu is a member of that group

**Task 4 — Permission Analysis**




**1,Owner permissions, Group permissions, Others permissions, and file type meaning.**

**In -rw-r--r--, the first - means it is a regular file. The owner has read and write permission, while the group and others have read-only permission**

**Task 5 — Numeric Permission Conversion**

**Convert rwxrwxrwx, rwxr-xr-x, rw-r--r--, rw-------, r-xr-xr-x to numeric values.**

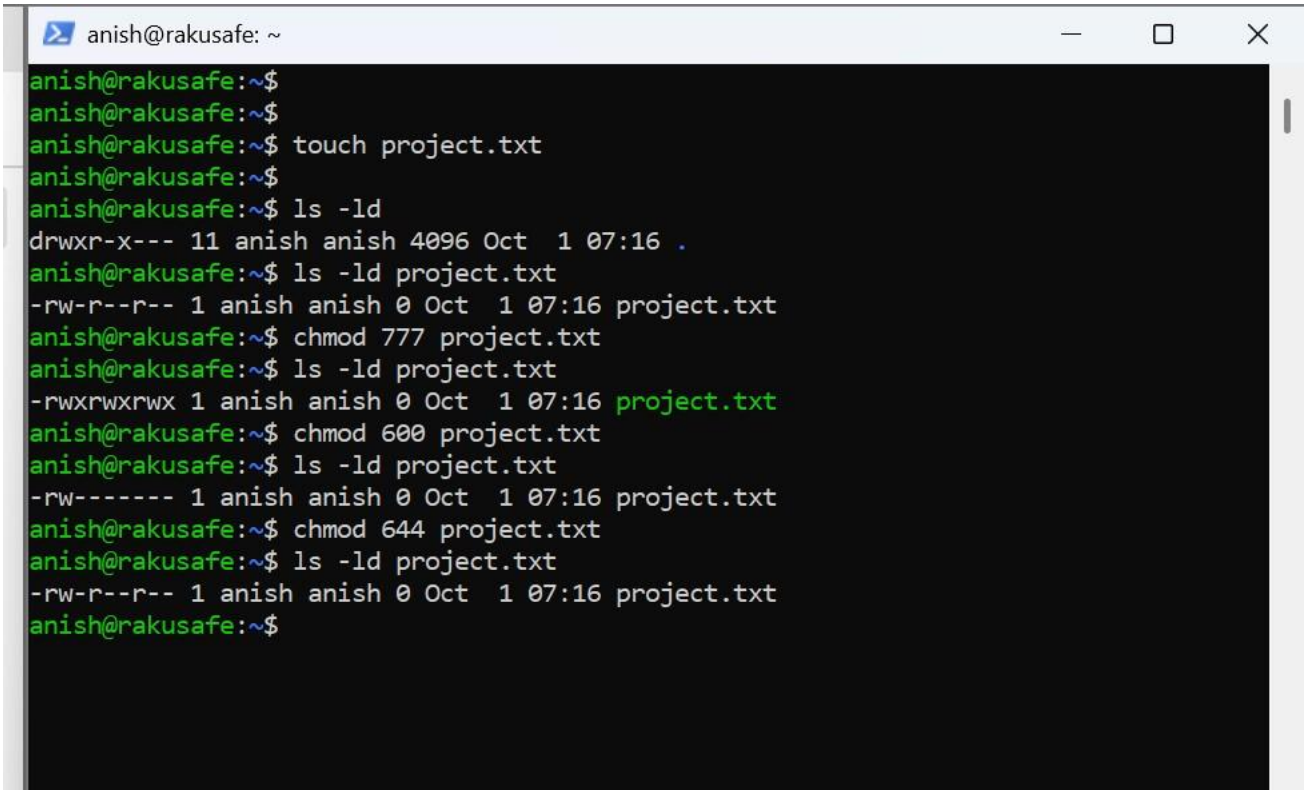
**Explain why 777 is dangerous and why 600 is recommended.**

| Symbolic Permission | Owner | Group | Others | Numeric |
|---|---|---|---|---|
| rwxrwxrwx | 7 | 7 | 7 | **777** |
| rwxr-xr-x | 7 | 5 | 5 | **755** |
| rw-r--r-- | 6 | 4 | 4 | **644** |
| rw------- | 6 | 0 | 0 | **600** |
| r-xr-xr-x | 5 | 5 | 5 | **555** |

**777 gives read, write, and execute permissions to everyone, so it creates a security risk. 600 allows only the owner to read and write the file, which protects sensitive data from unauthorized access.**

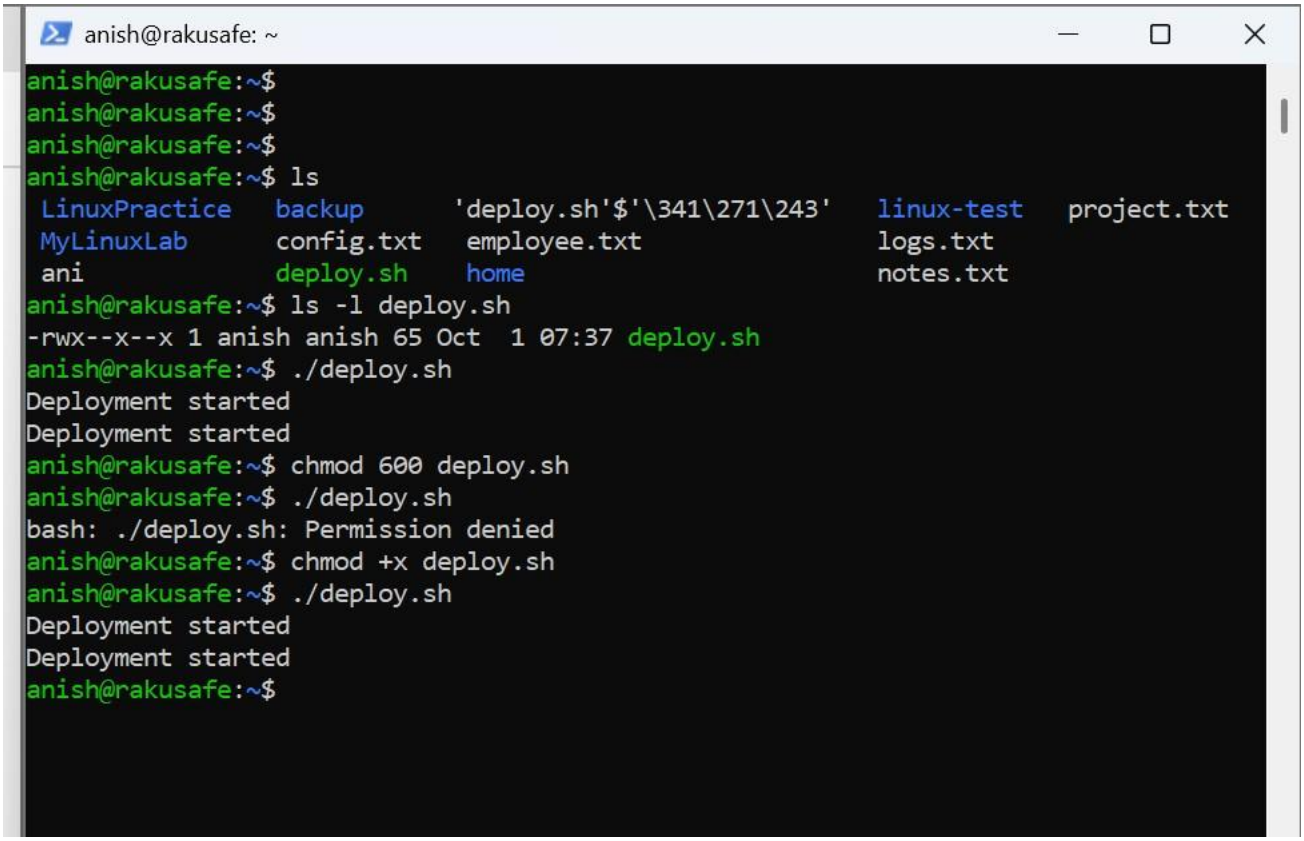
- **r (read) = 4**
- **w (write) = 2**
- **x (execute) = 1**
- **- (no permission) = 0**

**Task 6 — chmod Practical**



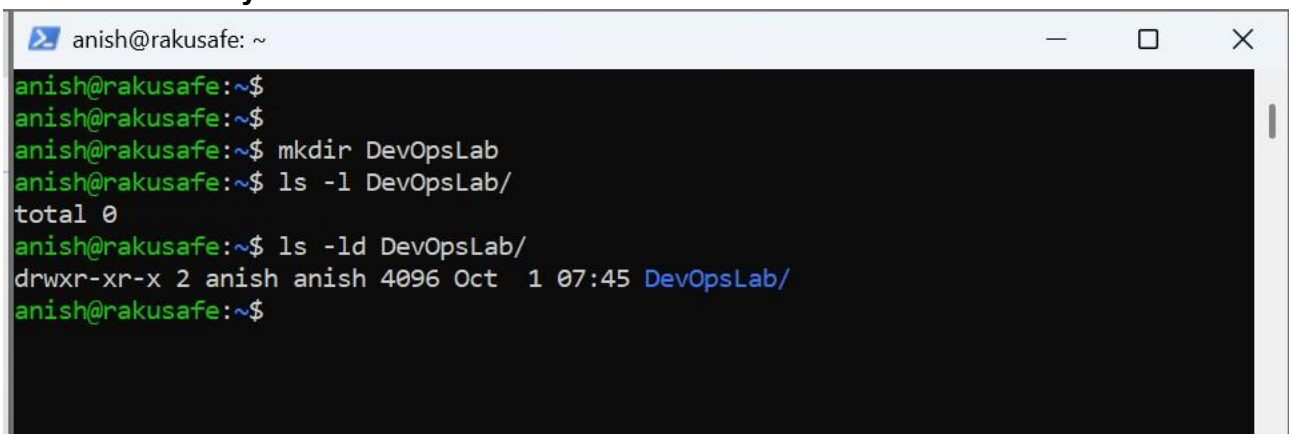

**chmod 777 gives full permissions to everyone, 644 gives read/write to the owner and read-only to group and others, while 600 gives read/write permission only to the owner.**

**Task 7 — Script Execution**




**chmod +x adds execute permission to a script, allowing it to be run directly. In DevOps, executable shell scripts are commonly used to automate deployment and system administration tasks.**

**Task 8 — Directory Permissions**

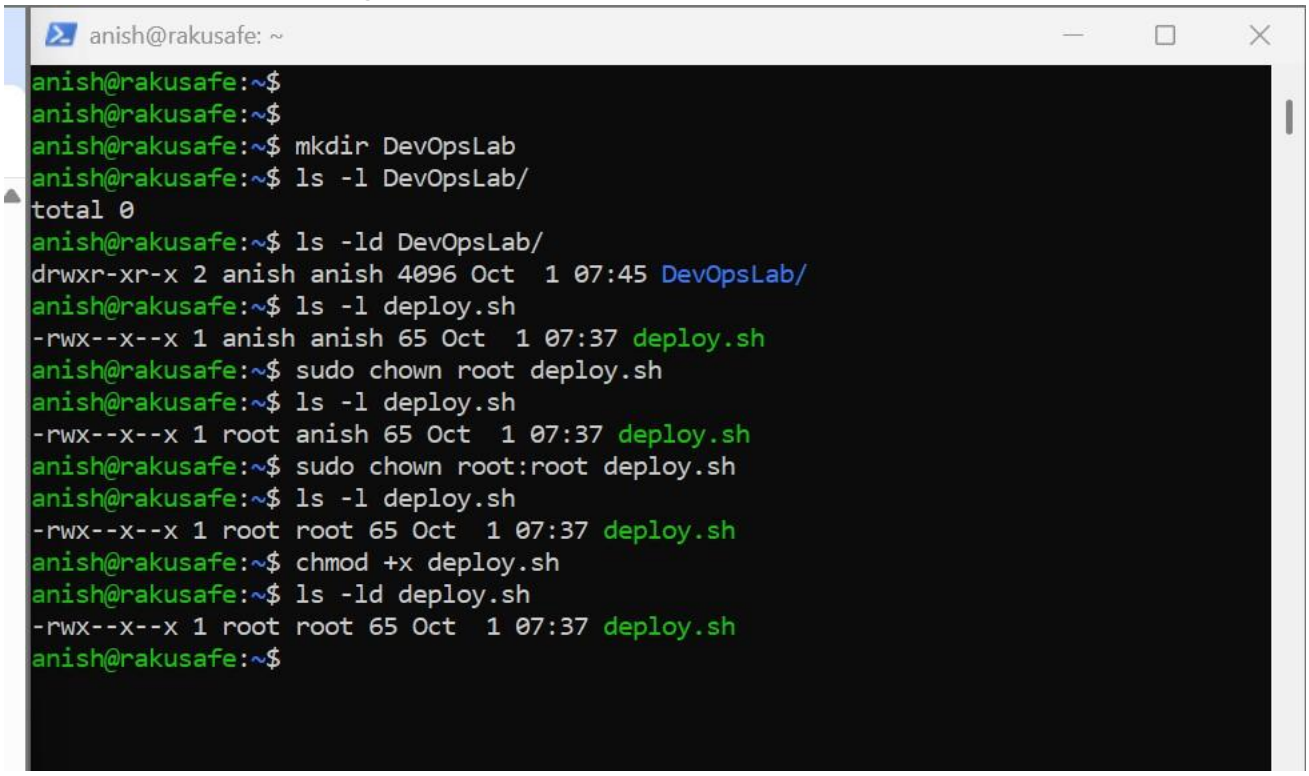



**Explain ls -l vs ls -ld and directory execute permission.**

**ls -l → contents of directory**
**ls -ld → directory itself**

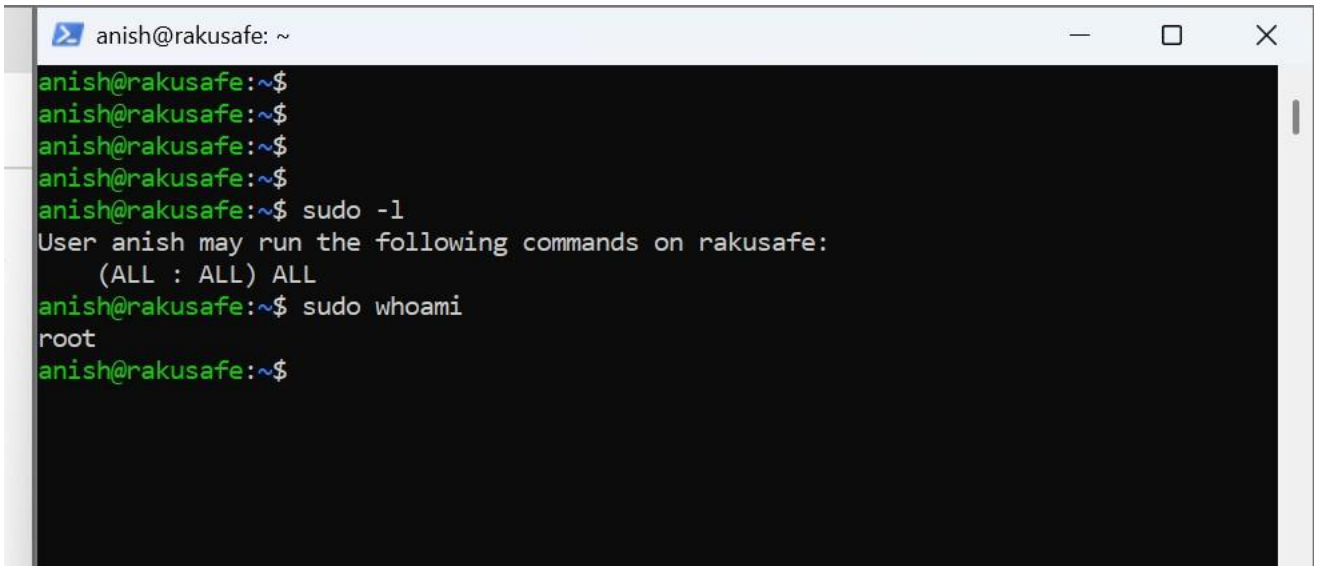
**For a directory, read permission allows listing its contents, write permission allows creating or deleting entries, and execute permission allows entering the directory and accessing its contents.**

**Task 9 — Ownership Management**




**chmod is used to change file or directory permissions, while chown is used to change the owner or group ownership**

**Task 10 — Sudo Verification**




sudo whoami
Output:root
Explanation: sudo runs the command with root (administrator) privileges. Therefore, whoami reports root.
sudo -l
This shows which commands your current user is allowed to run with sudo.
sudo = SuperUser DO
It allows a normal user to execute specific administrative commands with elevated/root privileges.

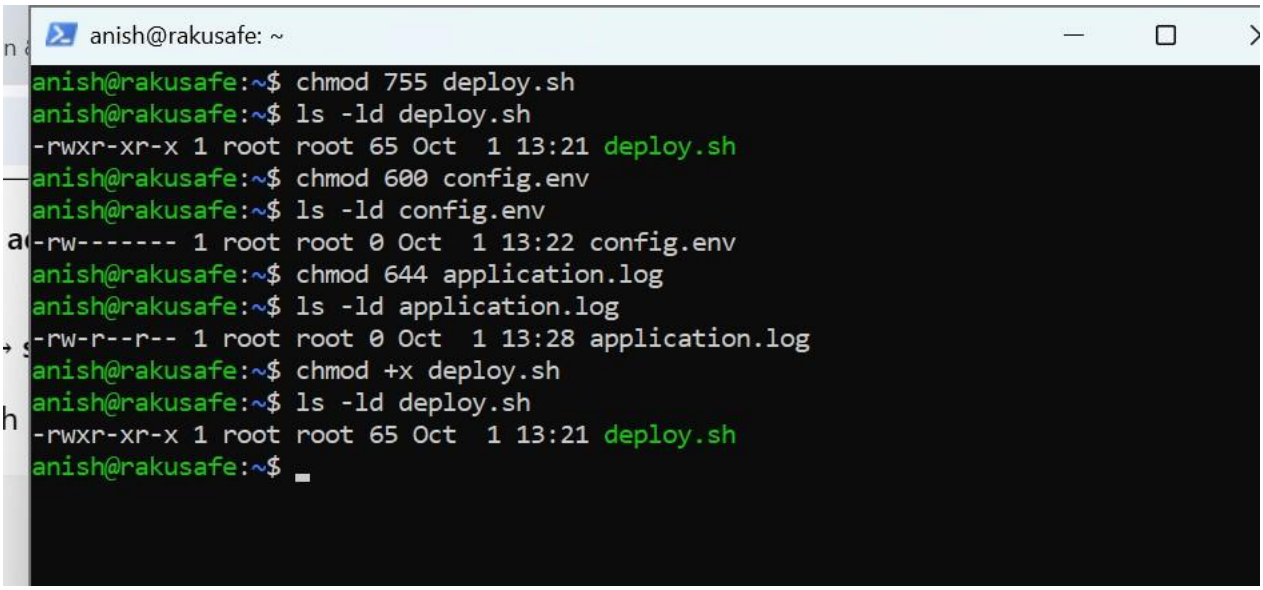
**Real DevOps Scenario**

**deploy.sh ® executable**

**config.env ® owner only**

**application.log ® owner write, others read**

**Assign correct permissions and justify**




**BONUS CHALLENGE**

